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Hydrocyclones as cell retention device for CHO perfusion processes in single-use bioreactors

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Abstract

In this study, a hydrocyclone (HC) especially designed for mammalian cell separation was applied for the separation of Chinese hamster ovary cells. The effect of key features on the separation efficiency, such as type of pumphead in the peristaltic feed pump, use of an auxiliary pump to control the perfusate flow rate, and tubing size in the recirculation loop were evaluated in batch separation tests. Based on these preliminary batch tests, the HC was then integrated to 50-L disposable bioreactor bags. Three perfusion runs were performed, including one where perfusion was started from a low-viability late fed-batch culture, and viability was restored. The successive runs allowed optimization of the HC-bag configuration, and cultivations with 20–25 days duration at cell concentrations up to 50×10^6 cells/ml were performed. Separation efficiencies up to 96% were achieved at pressure drops up to 2.5 bar, with no issues of product retention. To our knowledge, this is the first report in literature of high cell densities obtained with a HC integrated to a disposable perfusion bioreactor.

KEYWORDS

CHO cells, hydrocyclone, perfusion, separation efficiency, single-use bioreactor

1 | INTRODUCTION

Continuous cultivations with cell retention, also known as perfusion processes, have been increasingly developed by biopharmaceutical companies for manufacturing of therapeutic proteins, driven by needs for cost reduction, competition from off-patent products and compatibility to continuous downstream processing (Castilho, 2015; Karst, Steinebach, & Morbidelli, 2018; Sieck, Schild, & von Hagen, 2017; Walker, 2017). In past decades, perfusion had remained almost completely confined to the production of unstable proteins, and this was usually attributed to the higher technical complexity, the limitations of the cell retention devices then available, and the more dilute harvest due to continuous medium exchange. The advances in single-use technology, the development of more appropriate cell retention devices, and the progress in culture media and feeding strategies (Gagnon, Nagre, Wang, & Hiller, 2018; Konstantinov et al., 2006; Sieck et al., 2017), which allowed cell densities

in the order of 10^8 cells per milliliter (Clincke et al., 2013) and high product titers to be attained, were decisive in turning perfusion into the current mode of operation of choice, also for stable proteins.

Retention devices based on microfiltration systems and cell sedimentation, and externally connected to the bioreactor, are most frequently used for perfusion applications (Castilho, 2015; Chotteau, 2015; Woodside, Bowen, & Piret, 1998). Although microfiltration systems, such as alternating tangential flow (ATF) and tangential-flow filtration (TFF) have been extensively used, they are susceptible to fouling and clogging, leading to severe product retention (Karst, Serra, Villiger, Soos, & Morbidelli, 2016; Mercille, Johnson, Lemieux, & Massie, 1994; Wang et al., 2017) and/or premature interruption of the run (Clincke et al., 2013). Moreover, due to the complete cell retention capacity in such filtration-based systems, active bleeding is necessary to remove debris and dead cells from the bioreactor to maintain a healthy culture (Dalm, Cuijten, van

Grunsvén, Trämper, & Martens, 2004; Hiller, Clark, & Blanch, 1993). On the other hand, the low sedimentation velocity, density similar to water and high sensitivity to shear stress inherent to animal cells pose a challenge for separators lacking of a physical barrier, such as inclined settlers, acoustic separators, centrifuges, and hydrocyclones (Castilho & Medronho, 2002; Voisard, Meuwly, Ruffieux, Baer, & Kadouri, 2003; Woodside et al., 1998).

The hydrocyclone (HC) is a promising technology for separation in biological applications (Luebbert, Medronho, & Anspach, 2000; Ortega-Rivas, 2004; Rickwood, Onions, Bendixen, & Smyth, 1992; Yu & Fu, 2019). Its structure, comprising a cylindrical body and a conical portion, enables separation in a centrifugal field provided by the tangential inflow of the feed suspension at high flow rates. Besides being very compact, the absence of rotors or moving parts simplifies HC manufacturing and enables maintenance of performance in long-term operation (Castilho, 2015). Once the suspension is tangentially fed into the cylindrical section, an outer vortex enriched by denser and/or larger particles flows toward the underflow. The small underflow orifice size hinders the complete discharge, and part of the flow reverses into an inner secondary vortex that moves upward to leave the HC through the overflow pipe, conveying the smaller or lighter particles. As the cross-sectional area is reduced along the conical portion of the HC body, pressure drop (ΔP) inside the HC increases, determining the flow through the overflow orifice. An axial gas core is established in the HC as gas is aspirated through the underflow, forcing the fluid and less dense particles to leave through the overflow nozzle (Davailles, Climent, & Bourgeois, 2012).

The lack of a rigid cell wall is probably the main concern regarding the high shear forces that animal cells might be exposed to in a HC, especially due to the fact that the feed suspension is introduced at very high flow rates. However, separation tests with blood cells using a 15-mm diameter HC showed no evidence of cell lysis (Rickwood et al., 1992). No activation of apoptotic and necrotic cell death mechanisms for 48 hr after one HC passage was reported (Pinto, Medronho, & Castilho, 2008).

In the last two decades, several publications have emphasized the potential of HCs for animal cell separation (US Patent No. 6,878,545 B2, 2005; Deckwer, Medronho, Anspach, & Luebbert, 2005; Elsayed, Medronho, Wagner, & Deckwer, 2006; Jockwer, Medronho, Wagner, Anspach, & Deckwer, 2001; Luebbert et al., 2000; Medronho, Schuetze, & Deckwer, 2005; Pinto et al., 2008; Schröder, Elsayed, Olowina, & Wagner, 2010). Characteristics such as a small diameter (10 mm) of the cylindrical part, double-feed inlet, and optimization of the internal dimensions were considered to design HCs especially suited for mammalian cell separation (US Patent No. 6,878,545 B2, 2005; Deckwer et al., 2005). Despite the high retention efficiencies reported in these previous works (up to 99%), the rather low cell concentrations and eventual use of serum in media do not represent the current status of perfusion processes, which are nowadays carried out in serum-free media and reach tens or hundreds of millions of cells per milliliter. Moreover, the existing reports have not evaluated the long-term operation of the HC integrated into perfusion bioreactors.

In the present work, the performance of an HC was investigated in high cell-density perfusion cultures of mAb-producing Chinese hamster

ovary (CHO) cells in a 50-L disposable bioreactor. Before the final perfusion runs were carried out, preliminary batch tests were conducted to determine the effects on the separation efficiency of key operational conditions, such as type of feed pump, the possibility of controlling the overflow flow rate with a pump, and influence of the connectors used to attach the HC to the bioreactor bag.

2 | MATERIALS AND METHODS

2.1 | Cell lines and media

Suspension-adapted CHO cell lines were cultivated in chemically-defined animal component-free media at 37°C and 5–7.5% CO₂ atmosphere. The CHO DP-12 (ATCC #1934) cell line was cultivated in the TC-LECC medium (Xell AG, Germany) supplemented with 8 mM of glutamine. A CHO-K1 derived cell line (Cobra, Sweden) producing a recombinant monoclonal antibody (immunoglobulin G1 [IgG1]), here called CHO_mAb#1, was cultured in HyClone™ ActiPro™ medium (GE Healthcare) and used in perfusion cultivation #1. Another CHO-K1 derived cell line expressing an IgG antibody, here called CHO_mAb#2, was also cultured in ActiPro and used in perfusion cultivations #2 and #3. Cell density, viability, and cell size were determined by the trypan blue exclusion method using Vi-Cell XR (Beckman-Coulter) automated counter.

All the bioreactor cultivations were initiated with ActiPro as a basal medium. In run #1, ActiPro was also used as feed medium. In run #2, perfusion started with ActiPro, and then switched to ActiPro supplemented with Cell Boost™ 7a (7% vol/vol) and 7b (0.7% vol/vol; GE Healthcare). In run #3, ActiPro spiked with Cell Boost 1 (13.9% vol/vol) and 3 (15.8% vol/vol) was used as feed medium in the whole cultivation.

2.2 | Chromatographic beads to simulate cell particles

Chromatographic Superose 6 agarose beads (GE Healthcare, Sweden) with an average size of $12.6 \pm 0.3 \mu\text{m}$ and a relative density close to 1 were used in batch separation tests to mimic high-density cell suspensions. The beads could be reused since no damage was observed in the microscope after the HC passages. Bead suspensions were prepared with purified water at a concentration range between 5×10^6 and $150 \times 10^6/\text{ml}$ and kept under magnetic agitation at room temperature. Particle concentration and average diameter were measured using the Vi-Cell counter.

2.3 | Batch separation tests

Batch separation tests were first carried out using a stainless-steel HC (model HC2015) to evaluate operational features and alternatives for the attachment of the HC to single-use bioreactors bags. The HC2015 has a double tangential inlet, a 10-mm cylindrical

diameter and a conical angle of 6°. The number “2015” indicates an underflow diameter of 2.0 mm and an overflow diameter of 1.5 mm.

For these tests, bead or cell suspensions were fed to the HC with a peristaltic pump (Watson-Marlow 520U, UK). Pressure drops (ΔP) between 0.5 and 3.0 bar were established by increasing the rotational pump speed, and measured by an oil–liquid manometer (Wika, Brazil) or a disposable pressure sensor (SciPress™, Parker Hannifin) placed close to the bifurcation to the feed inlets, as shown in Figure 1a. Samples from underflow and overflow were collected at timed intervals in pre-tared beakers for the determination of flow rates, particle concentration and, in case of experiments with cells, cell viability. Flow rates and particle concentrations were determined in duplicate for each experimental point. All tests were performed at room temperature.

2.3.1 | Comparison of high- and low-pulsation feed pumps

Tests with a conventional pulsating (520R2) and a low-pulsation (505L) Watson-Marlow pumpheads compatible to the 520U peristaltic pump were performed with cell and bead suspensions. For this comparison, a Marprene tube with 8.0 mm internal diameter (#902.0080.024; Watson-Marlow) was mounted in the pumpheads.

For the 505L pumphead, which has a double track, Y-tubing elements were fitted on each side of the pumphead, according to manufacturer's instructions. CHODP-12 cell suspensions diluted in phosphate-buffered saline, or bead suspensions at a concentration range of 30×10^6 – 150×10^6 /ml, were fed into the HC at pressure drops between 0.5 and 1.5 bar.

2.3.2 | Control of overflow flow rate with a peristaltic pump

A second 520U peristaltic pump with a 520R2 conventional pulsating pumphead was installed in the overflow line with a Marprene tubing of 3.2 mm internal diameter (#902.0032.024; Watson-Marlow) and switched on simultaneously with the HC feed pump (Figure 1a). Bead suspensions at approximately 5×10^6 and 50×10^6 /ml were fed at pressure drops between 1 and 2 bar.

2.3.3 | Type of connection in the recirculation loop

At first, batch tests were carried out to evaluate the effect of two elements to be used in the underflow-bioreactor connection

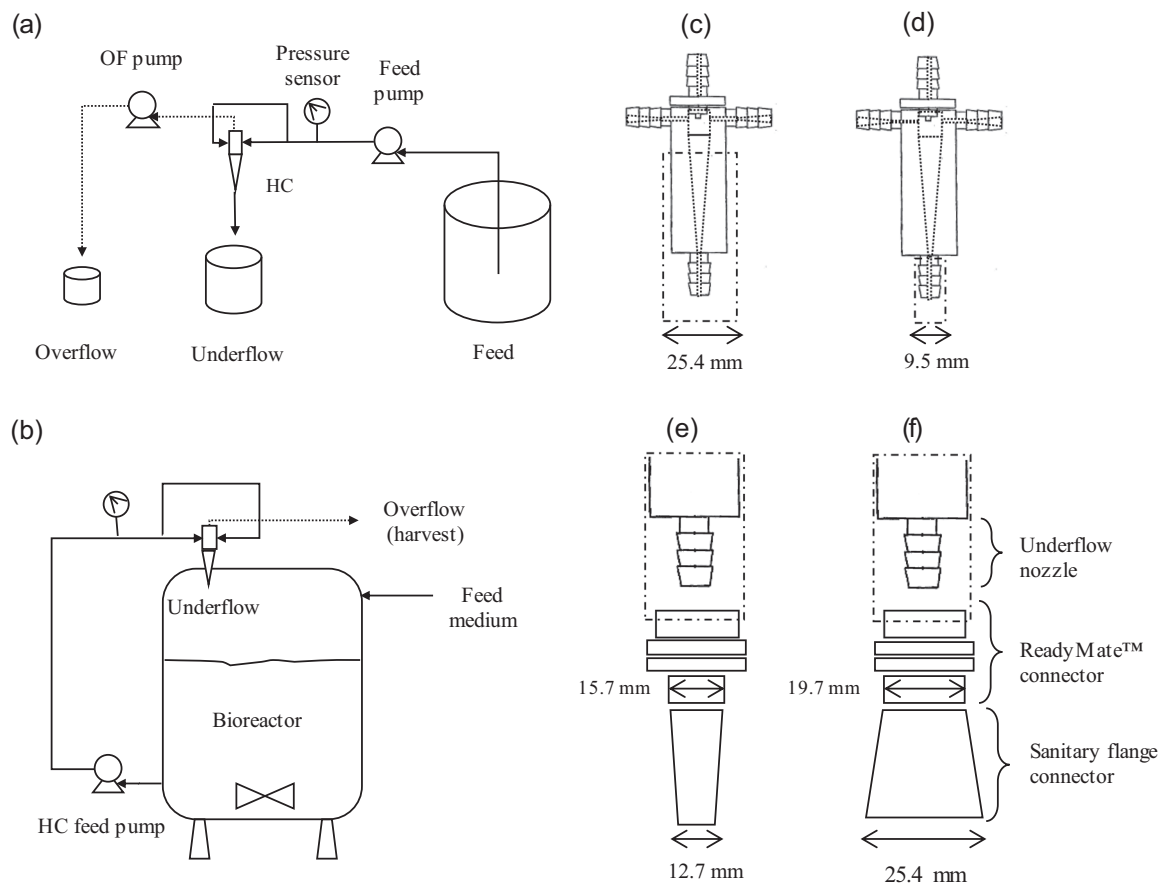


FIGURE 1 Schematic drawings of (a) test rig for batch tests with the hydrocyclone (HC), including an auxiliary pump at the overflow (OF) line for Q_{of} control tests; (b) proposition for the HC attachment to a single-use perfusion bioreactor. On the upper right, the HC with extended underflow outlet with (c) 25.4-mm and (d) 9.5-mm ID tubing. ReadyMate connectors with (e) 12.7-mm ID and (f) 25.4-mm sanitary flanges positioned at the underflow outlet. Dotted lines represent silicone tubings. ID, internal diameter

(recirculation loop) on separation efficiency: tubing diameter and presence of ReadyMate™ Tri-Clamp Disposable Aseptic Connectors of different sizes (GE Healthcare). In a first approach, silicone tubings having 25.4 mm (1") or 9.5 mm (3/8") internal diameter (ID) were attached either to the HC cylindrical body or to the underflow nozzle (Figure 1c,d). A minimum tubing length of 30 cm was used in case that the recirculation loop would be directly provided by tubing weld or fast-connectors available in commercial bags. In a second approach, ReadyMate connectors were joined, giving shorter lengths in the underflow-bioreactor loop. ReadyMate connectors with 15.7 mm ID (#28-9366-95; Mini TC) and 19.7 mm ID (#28-9568-89; Standard TC) were attached to the HC body with silicone tubings and connected respectively to 12.7-mm and 25.4-mm sanitary flanges with barb (Figure 1e,f). Both assemblies were mounted on the top of a 10-L Schott flask, and CHO cells were fed at pressure drops between 1 and 3 bar.

2.3.4 | Determination of separation efficiency

The total separation efficiency (E_T) was calculated as the fraction of the number of feed particles recovered in the underflow (Equation 1; Castilho & Medronho, 2008):

$$E_T = \frac{Q_{uf} N_{uf}}{Q_f N_f}, \quad (1)$$

where Q_f and Q_{uf} are the feed and underflow flow rates, and N_f and N_{uf} are the particle concentrations (beads or viable cells per unit volume) in the feed and underflow, respectively.

The flow ratio (R_f), which represents the minimum separation efficiency related to the drag force (Castilho & Medronho, 2008), was calculated according to Equation 2:

$$R_f = \frac{Q_{uf}(1 - C_{vu})}{Q_f(1 - C_{vf})}, \quad (2)$$

where C_{vu} and C_{vf} are the volumetric concentrations (vol/vol) of solids in the underflow and feed, respectively. For the cell concentration ranges used in this work, R_f is approximately equal to Q_{uf}/Q_f .

The net centrifugal separation efficiency (E' , also known as "reduced efficiency") was calculated according to Equation 3 (Castilho & Medronho, 2002):

$$E' = \frac{E_T - R_f}{1 - R_f} \approx 1 - \frac{VCD_{of}}{VCD_f}, \quad (3)$$

where VCD_f and VCD_{of} are the viable cell concentrations in the feed and overflow, respectively.

2.4 | Perfusion cultivations

The stainless-steel HC2015 prototype was installed on the top of gamma-sterilized commercial bioreactor bags, as schematically shown in Figure 1b. Three perfusion cultivations with two mAb-producing CHO cell lines were performed in a 50-liter system (Xcellerex™ XDR-50, GE Healthcare) with 40 liters of working volume.

Temperature (36.8°C), pH (7.10), dissolved oxygen (DO, 40% of air saturation), and agitation were monitored and controlled using a X-station/Wonderware software system (GE Healthcare). Air and pure O₂ for DO control, and CO₂ for pH control were supplied through a microsparger, whereas a continuous airflow at 1–1.5 L/min was injected through a 1-mm macrosparger for CO₂ stripping. Sodium bicarbonate solution (7.5% m/vol) was used for pH control, and antifoam-C (Sigma-Aldrich) at 3 % (vol/vol) was added as required. Samples from the bioreactor and the overflow were daily withdrawn for determination of cell density and viability, metabolites (BioProfile FLEX, Nova Biomedical), IgG titer, and lactate dehydrogenase (LDH) activity (Cedex Bio Analyzer, Roche, Germany).

The wholecell retention assembly, that is, HC tubing connections with braided silicone hoses, a SciPress™ disposable pressure sensor and, if available, a single-use rotary flowmeter (FM-23WV; PendoTECH), was autoclaved and installed in the bioreactor bag according to the available connectors (ReadyMate ports and weldable thermoplastic tubings, described in the text). The overflow tubing was connected to a 200-L harvest bag.

In run #1, the HC was installed in a 50-L bag (XDR-50 DEV, #888-0356-C; GE Healthcare) by welding a 9.5-mm ID thermoplastic tubing between the underflow and the bioreactor. For runs #2 and #3, customized bag models #888-0086-C (PRO) and #888-0356-C (DEV) fitted with a 19.7 mm ID ReadyMate TC connector welded on the top of the bag were respectively used. The main differences between both bioreactor bag models consisted of the number of ports, tubing sizes available for auxiliary flasks (e.g., base, feed line, etc.) and microsparger pore size (2 µm for PRO, 20 µm for DEV). The HC feed pump was intermittently activated using a lamp timer (Diehl 881-4, Germany), to stay on for 15 min once or more times per day, according to the desired perfusion rate. Every time the HC was operated, partial removal of the cell broth through the overflow simultaneously activated the medium feed pump, to maintain the bioreactor volume constant.

2.4.1 | Evaluation of product quality

Product quality attributes were determined for perfusion runs #2 and #3 performed with CHO_mAb#2 cell line. UPLC analyses to determine N-glycans (GlycoWorks™ RapiFluor-MS N-Glycan Kit; Waters) and aggregates (Superdex™ 200 Increase 5/150 GL; GE Healthcare, Sweden) were carried out according to manufacturers' instructions in bioreactor samples, which were collected at early and late stages of the perfusion runs and purified with protein-A resin (HiTrap™ Protein A HP; GE Healthcare, Sweden).

3 | RESULTS AND DISCUSSION

3.1 | Effect of low- and high-pulsation pumpheads on separation efficiency

Separation tests with suspensions of CHO cells and Superose 6 beads were performed to determine whether the intense pulsation promoted

by a conventional peristaltic pumphead (Watson-Marlow 520R2) could negatively affect HC performance. Previous reports using HCs for animal cell separation had used a low-pulsation pumphead (Watson-Marlow 505L) in the feed pump (Elsayed et al., 2006; Jockwer et al., 2001; Pinto et al., 2008; Schröder et al., 2010), to minimize eventual effects on the flow pattern inside the HC. This pumphead has six low-inertia stainless-steel rollers and twin offset tracks that yield a smooth, low-pulsation flow.

To correlate pump rotational velocities and ΔP in the HC, preliminary tests with water with the low-pulsation pumphead, using either 8.0 or 9.6-mm tubing bores, gave a maximum ΔP of 1.6 bar at the maximum rotational speed (220 rpm). This value is in agreement with Elsayed et al. (2006), who determined a maximum pressure drop of 1.4 bar for the same 505L model using HCs of two different geometries.

Separation tests were first carried out using a CHO DP-12 cell suspension at approximately 8×10^6 cells/ml and viability over 90%. Similar values for feed flow rate (Q_f) were obtained with both pumpheads (Table 1), whereas slightly higher Q_f values were previously determined for the same high-pulse pumphead (Elsayed, Ramalho, Castilho, & Medronho, 2012). Moreover, the low-pulsation pumphead demanded higher rotational velocities to yield the same ΔP compared with the conventional pulsating one. A constantflow ratio (R_f) value of 77% was determined for the whole ΔP range tested, which is in agreement with Pinto et al. (2008) (77.4%) and Jockwer et al. (2001) (75.3%) for the same HC geometry (HC2015). Cells were concentrated in the underflow up to 1.3-fold in both pumpheads, similarly to the concentration factor reported for BHK and HEK cells separated with a HC of similar geometry (HC2520; Elsayed et al., 2006). Viability of cells recovered from the underflow in our tests decreased up to 1.1% compared with the feed suspension (Table 1, see ΔV_u), as compared with decreases of 9–11% reported in previous works (Jockwer et al., 2001; Pinto et al., 2008). Overall, slightly higher separation efficiencies were observed in our tests for the 505L pumphead for every pressure drop, with a maximum total separation efficiency (E_T) close to 96%. As the centrifugal field became more intense at higher ΔP , centrifugal separation efficiencies (E') increased up to 77%. Higher E' values correlated with more diluted overflow suspensions, with viable cell concentrations in the overflow decreasing from 3.7×10^6 to 1.8×10^6 cells/ml using the 520R2 pumphead (Table 1).

A more comprehensive experimental plan was carried out with Superose 6 beads as model particles with properties close to cells, at concentrations between 30×10^6 and 150×10^6 /ml. As previously observed in the separation tests with CHO cells, for a given ΔP a higher rotational velocity was required for the low-pulsation pumphead as compared with the conventional pulsating one, whereas a R_f value of 77% was determined for both pumpheads at all pressures drops. No significant differences on separation efficiency were observed (Figure 2). Differences due to the pumphead type were only observed for Q_f measured at 0.5 bar at all feed concentrations (Figure 2a). These data is in agreement to Elsayed et al. (2012), who found no differences in separation efficiency for both pumpheads when testing a similar hydrocyclone (HC2010).

Based on the average bead diameter determined using the Vi-Cell counter ($12.6 \pm 0.3 \mu\text{m}$), the volumetric concentrations fed to the HC were in the range of 3–15% (vol/vol). According to the data shown in Figure 2, no effect of feed concentration was observed on separation efficiency for this range of solid fraction. However, when compared to the separation efficiencies obtained for CHO cells (Table 1), the HC performance was significantly lower for the beads, which is probably related to the slightly smaller size of the beads. The terminal settling velocity of a particle under the Stoke's region increases with the square of its diameter, so slight differences in size can cause significant effects on separation efficiency.

The conventional pulsating peristaltic pumphead (520R2) was chosen to continue the studies on the HC for different reasons: it gave a similar performance as compared with the low-pulsation pumphead (505L) in terms of separation efficiency, Q_f , R_f , and cell viability; it can achieve higher pressure drops for a given pump speed than the 505L pumphead, enabling tests at a broader pressure range and potentially enabling better performance; it is the conventional type of peristaltic pumphead usually available in process development laboratories. Thus, using CHO_mAb #2, the conventional pulsating pumphead was further tested up to pressures drops of 2.9 bar (Table 1). When ΔP increased from 1.2 to 2.9 bar, R_f remained constant at 78%, whereas E_T increased up to 97% and centrifugal efficiency experienced a significant increase from 71% to 85% at 2.9 bar, resulting in a more clarified harvest discharged through the overflow (1.5×10^6 at 1.2 bar as compared with 0.5×10^6 cells/ml at 2.9 bar). However, cell viability in the underflow (ΔV_u) decreased 2.4–3.4% when ΔP was 2.7–2.9 bar. A much higher decrease of 14% in the underflow viability was determined by Pinto et al. (2008) when operating a HC2020 at 3 bar. In contrast, the viability of HeLa cells concentrated in the underflow did not significantly decrease when separated at 4 bar in a commercial Dorr-Oliver HC (Luebbert et al., 2000). Based on our present results shown in Table 1, despite the gain in separation efficiency provided at higher ΔP , the effects on cell viability could become more relevant, especially in case of continuous operation of the HC in a perfusion process. Therefore, ΔP was selected as an important variable to investigate in perfusion runs in this work. Empirical models correlating the applied pump velocities with ΔP , Q_f , and Q_{of} based on these batch test results were proposed (Figure S1). Furthermore, in these experiments, during the batch operation of the HC2015 at open atmosphere, some features related to the fluid swirling motion inside the HC were confirmed. The underflow discharge had a typical spray or umbrella shape, whereas the formation of an air-core inside the HC could be confirmed by the presence of air bubbles released through the overflow.

3.2 | Control of perfusate flow rate with an auxiliary peristaltic pump in the overflow line

The HC2015 used in this work can process up to 0.5 L/min of perfusate at 2.9 bar through the overflow orifice (Table 1). This high

TABLE 1 Effect of pulsating and low-pulsation pumpheads on the separation efficiency of CHO cells for hydrocyclone HC2015 at pressure drops between 1 and 3 bar. ΔV_u corresponds to cell viability in the feed minus viability in the underflow, as defined by Pinto et al. (2008). Due to the limited volume of feed cell suspension, and to avoid misinterpretations in the assessment of cell viability, CHO DP-12 cells were passaged once through the hydrocyclone. In this case, no replicates were performed for each experimental condition

ID pump tubing (mm)	Cell line	Pumphead type	Pressure (bar)	Pump speed (rpm)	Q_f (L/min)	Q_{of} (L/min)	R_f (%)	VCD_f ($\times 10^6$ cells/ml)	VCD_{ur} ($\times 10^6$ cells/ml)	VCD_{of} ($\times 10^6$ cells/ml)	E_T (%)	E' (%)	ΔV_u (%)
8.0	CHO DP-12	Low-pulsation (505 L)	0.5	135	1.17	0.27	76.5	8.68	10.11	3.54	90.6	59.2	0.06
			1.0	162	1.40	0.32	76.9	8.53	10.73	1.65	95.6	80.6	0.71
			1.5	200	1.57	0.35	77.8	8.29	10.72	1.93	95.1	76.8	1.10
			0.5	100	1.18	0.28	76.0	7.41	10.01	3.67	89.6	50.5	-0.25
			1.0	141	1.45	0.34	76.4	7.69	9.67	2.07	93.8	73.1	0.15
9.6	CHO_mAb#2	Pulsating (520R2)	1.5	185	1.61	0.38	76.7	7.27	9.83	1.84	94.6	74.7	1.05
			1.2	83	1.47 $\pm$ 0.04	0.33 $\pm$ 0.04	78.0 $\pm$ 2.2	5.08	6.22 $\pm$ 0.31	1.49 $\pm$ 0.40	93.8 $\pm$ 0.6	71.5 $\pm$ 5.4	-0.20 $\pm$ 0.29
			1.7	100	1.83 $\pm$ 0.06	0.40 $\pm$ 0.01	78.1 $\pm$ 0.3	4.80	5.44 $\pm$ 0.23	0.97 $\pm$ 0.17	95.3 $\pm$ 0.9	78.3 $\pm$ 4.4	-0.46 $\pm$ 0.19
			2.1	120	1.95 $\pm$ 0.02	0.43 $\pm$ 0.01	78.0 $\pm$ 0.2	4.32	5.34 $\pm$ 0.29	0.87 $\pm$ 0.09	95.6 $\pm$ 0.7	79.9 $\pm$ 3.0	0.30 $\pm$ 0.23
			2.4	140	2.06 $\pm$ 0.01	0.46 $\pm$ 0.01	77.7 $\pm$ 0.6	4.53	5.57 $\pm$ 0.32	0.89 $\pm$ 0.07	95.6 $\pm$ 0.1	80.3 $\pm$ 0.2	0.57 $\pm$ 0.01
			2.7	170	2.21 $\pm$ 0.05	0.49 $\pm$ 0.00	77.8 $\pm$ 0.4	4.24	4.92 $\pm$ 0.35	0.62 $\pm$ 0.11	96.5 $\pm$ 0.7	84.3 $\pm$ 3.6	2.39 $\pm$ 0.69
			2.9	220	2.30 $\pm$ 0.03	0.51 $\pm$ 0.01	77.9 $\pm$ 0.1	3.89	4.47 $\pm$ 0.13	0.54 $\pm$ 0.01	96.7 $\pm$ 0.1	85.0 $\pm$ 0.3	3.39 $\pm$ 0.11

Abbreviations: CHO, Chinese hamster ovary; ID, internal diameter; Q_f , feed flow rate; R_f , flow ratio; VCD, viable cell density; VCD_f , viable cell density in the feed; VCD_{ur} , viable cell density in the underflow; VCD_{of} , viable cell density in the overflow.

capacity would make a single and small 10-mm HC suitable for a 300-L bioreactor operated at a perfusion rate of approximately 2.4 reactor volumes per day (RV/day). For smaller vessel sizes or lower perfusion rates, this HC could be continuously operated at lower pressure drops to reduce Q_{of} . However, the lower ΔP could decrease process performance. Therefore, the intermittent operation of the HC at the desired ΔP is often adopted by switching on and off the feed pump (Elsayed & Wagner, 2011; Jockwer et al., 2001; Schröder et al., 2010). To evaluate if the perfusate flow rate (Q_{of}) could be controlled by means of an auxiliary pump in the overflow line, we compared HC operation with an uncontrolled overflow (control experiment) to overflow flow rates controlled by a peristaltic pump (520U/520R2) operating at either 0.040 L/min or 0.077 L/min, which would result in perfusion rates in the range of 1–2 RV/day for a 50-L bioreactor. Although similar Q_f values were obtained compared to the control condition (i.e., unrestricted overflow), R_f increased from 77% to 98% due to a preferential flow partition to the underflow orifice caused by the restriction in Q_{of} . Despite the increase of E_T up to 98% for both Q_{of} values with bead suspensions at 5×10^6 and 50×10^6 /ml (Figure S2), this total separation efficiency was almost solely a result of the division of streams to underflow and overflow, and not due to centrifugal action. As a result, bead concentrations measured in the overflow and feed suspension were approximately the same. This behavior was also related to the lack of the gas core inside the HC, since no air bubbles coming out from the overflow were observed, indicating that the pump restricting considerably (4.33–11.4 fold) the overflow flow rate negatively impacted the flow pattern inside the HC.

3.3 | Effect of diameter size in the HC-bioreactor recirculation loop

Next step before carrying out perfusion runs was to determine an adequate way to connect the HC to the bioreactor bag. A tubing with 9.5-mm ID would provide a straightforward connection between the HC and the bioreactor for returning the cells, since this is the outer diameter of the underflow nozzle. However, the size restriction imposed by a 9.5-mm tubing attached to the underflow significantly decreased E' , as shown in Figure 3a. The underflow orifice was promptly submerged by the underflow suspension, as the flow restriction hampered a faster flow at higher pressure drops. Additionally, the underflow discharge did not show an umbrella shape, and no air bubbles exiting from overflow were observed, indicating that the flow pattern inside the HC had been disturbed. Similarly to the test of a controlled Q_{of} using an auxiliary pump, a slightly higher E_T was obtained with the narrower 9.5-mm tubing, but this was due to an increase in R_f from 77% to 85%, that is, simply due to fluid partition and not to centrifugal action. On the other hand, a larger 25.4-mm tubing attached to the cylindrical HC body provided E_T values similar to the control condition (i.e., no tubing connecting the underflow to the bioreactor), did not alter R_f and preserved the umbrella-pattern discharge. Thus, using a

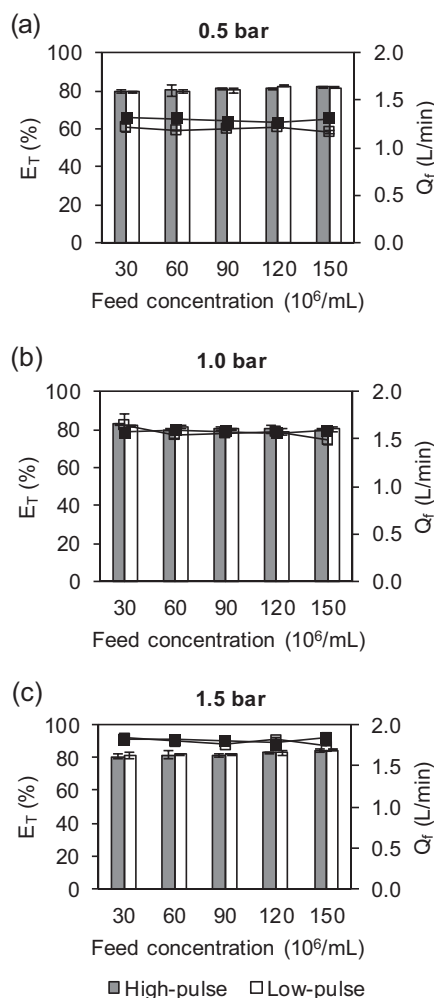


FIGURE 2 Effect of high- (closed) and low-pulsation (open symbols) pumphead type on E_T (bars) and feed flow rates (lines) for Superose 6 bead concentrations between 30×10^6 and 150×10^6 /ml at: (a) 0.5 bar, (b) 1.0 bar, and (c) 1.5 bar. Data shown as mean $\pm$ SD for $n = 2$. SD, standard deviation

25.4-mm silicone tubing attached to the HC body, ReadyMate connectors and sanitary flanges with minimum ID of 12.7 and 19.7-mm (Figure 1e,f) were tested to enable the connection of the HC to the bioreactor bag. As shown in Figure 3b, the narrower 12.7-mm reducer caused a significant decrease in E_T and E' at 1.9 bar, when the umbrella-pattern discharge in the underflow suspension was disrupted. Instead, a larger 19.7-mm connector enabled an umbrella-type discharge and gave a better performance up to the highest pressure drop tested (2.3 bar). A CHO cell suspension at 24×10^6 cells/ml was fed at 2.3 bar, resulting in a diluted overflow stream at 4.4×10^6 cells/ml, corresponding to a total separation efficiency (E_T) of 96% and a centrifugal efficiency (E') of 82%. To our knowledge, this was the first time that a HC was challenged with a feed suspension at a viable cell density higher than 9.8×10^6 cells/ml (Elsayed, 2005), and the connector configuration proposed herein enabled a separation performance comparable to low concentration feed suspensions.

3.4 | Perfusion runs

Perfusion cultivations were performed with HC2015 connected to a 50-L single-use bioreactor bag, and a 520U/520R2 conventional pulsating peristaltic pump with a 9.6-mm ID tubing was used to feed the HC. Due to the high Q_{of} rates provided by the unrestricted overflow, the HC was operated in an intermittent fashion to provide the desired perfusion rates.

Run #1, which consisted of a preliminary evaluation of the HC coupling to a commercial disposable bag, started by diluting a late-stage fed-batch CHO_mAb#1 culture obtained from a 13-day fed-batch process. The disposable bag used was a standard model, and the HC was connected to the bag in a nonoptimal way, using a 1-m long silicone tubing and a 12.7-mm sanitary flange reducer. On the day considered as Day 0 of this run, cells with viability lower than 70% were diluted to 40 L in the ActiPro medium supplemented with 6 mM glutamine. Perfusion was started on Day 3 by activating the HC feed pump to withdraw 10 liters per day through the overflow (i.e., applying a perfusion rate of 0.25 RV/day), followed by replenishment of the bioreactor volume with fresh medium. Perfusion rate was increased to 0.5 RV/day (medium exchange once a day) and kept constant throughout the cultivation (Figure 4b). The HC was operated at 1 bar. Approximately 10 min after switching on the HC pump, in spite of its length, the recirculation loop was completely filled with liquid, suppressing the umbrella-pattern discharge of the underflow, eliminating the gas core and disrupting the flow pattern inside the HC. Despite the nonideal HC configuration, cell densities of 14×10^6 cells/ml and viabilities over 95% were achieved (Figure 4a). Separation efficiencies related to the centrifugal action (E') were calculated based on cell concentration measurements in the bioreactor and in the overflow stream (Equation 3). E' around 70%, and E_T values between 91% and 95% were observed throughout the run (Figure 4b). E_T values were calculated based on $R_f = 82\%$, which was determined afterward in batch tests that imitated the submerged drowned underflow condition caused by the 12.7-mm connector.

In the following runs using CHO_mAb#2 cells, the submerged underflow issue was solved by customizing the bioreactor bag with a dedicated 1.5" ReadyMate port, to install the HC directly on the top of the bag. Thus, the underflow suspension was freely discharged inside the bioreactor, and the typical HC umbrella-type discharge was observed. Apart from the late-stage fed-batch culture used as inoculum and the nonideal HC connection in run #1, runs #2, and #3 were improved also by smoothing HC operation along the days activating the feed pump for 15 min every 3–6 hr, depending on Q_{of} (0.3–0.4 L/min, monitored by a scale—see Figures 5e and 6e) and the desired perfusion rates (20–40 L/day). With this simple strategy, the HC was reliably operated for up to 23 days without interruption. Antifoam was added only once a day, even when the highest perfusion rates were applied and more HC cycles per day were required, which potentially could have caused foam issues due to the turbulent discharge of the underflow suspension into the bioreactor.

Higher cell concentrations were achieved in runs #2 and #3 compared with run #1, as illustrated in Figures 5a and 6a. In run #2, basal

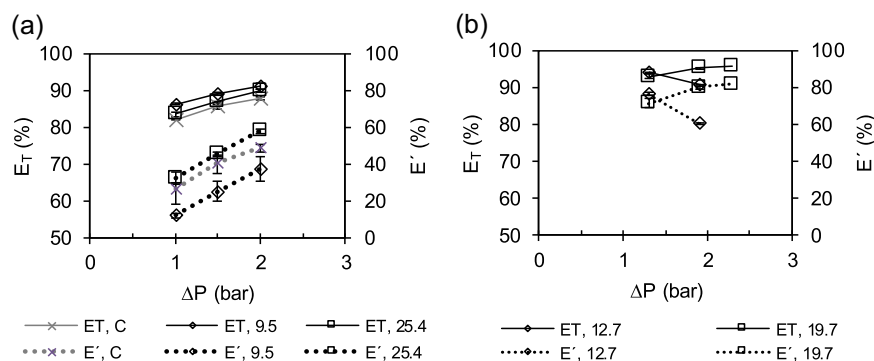


FIGURE 3 (a) Effect of tubing diameter size (9.5 vs. 25.4-mm ID) on separation of Superose 6 beads at 5×10^6 /ml; control (C) refers to an open underflow discharge without tubing. (b) Effect of connector size (12.7 vs. 19.7-mm ID) on separation of CHO cells at a concentration range of 15.3×10^6 – 23.7×10^6 cells/ml; data are shown as mean $\pm$ standard deviation (except for 25.4 mm, 2.3 bar, where $n = 1$). CHO, Chinese hamster ovary; ID, internal diameter

medium ActiPro was used as perfusion medium between Day 3 and 17, and then replaced with a fortified medium (Reinhart, Damjanovic, Castan, Ernst, & Kunert, 2018) to allow an increase of cell density in the bioreactor. However, differently from all previous experiments that used a 3/8" (9.6 mm) tubing to feed the HC, a double-Y tubing with 3.2 mm ID (1/8") was the only available option in the PRO bag model to connect the HC feed line. The smaller diameter of the feed line caused a significant pressure drop, requiring for a given flow rate pump velocities up to 58% higher than those previously used in batch tests. Therefore, the feed suspension entered at lower Q_f into the HC, leading to a lower pressure drop across the HC, and E' values were low ($52 \pm 6\%$). On Day 12, an extra 3.2-mm port originally designated as a sampling port was welded to the bifurcated feed line. The additional line allowed an increase in Q_f , and ΔP rose from 1.5 to 1.8 bar for the same pump velocity, resulting in a step increase in E' from 48% to 68%. Cell density in the overflow decreased from approximately 16×10^6 to about 10×10^6 cells/ml (Figure 5a). On Day 19, a single-use rotary flowmeter was aseptically installed upstream of the feed pump for an in-line Q_f measurement, consequently allowing Q_{uf} calculation. Thus, E_T values between 91.6% and 95.5% were determined (Figure 5c).

A sharp increase in LDH activity from Day 18 onwards initially suggested a negative impact of the fortified feed medium that had been introduced on Day 17 (Figure 5d). Due to the richer medium, perfusion rate was decreased and then gradually increased up to 1 RV/day (Figure 5a), achieving a condition of moderate hyperosmolality (up to 350 mOsm/kg). On the other hand, pressure drop in the HC was increased on Day 19 and kept at an average of 2.2 bar until the end of the run, to improve separation performance and to minimize the loss of cells through the overflow. This strategy required the application of the maximum pump velocity to compensate the pressure drop caused by the small diameter HC feed tubing. Thus, the high pump velocities may have also contributed to cell lysis and release of intracellular LDH, since as shown in Table 1 high pumps speed have been correlated with cell viability decrease. To minimize any possible negative impacts of pumping on cell growth and productivity, according to Chotteau (2015) larger tube diameters should be preferred to reduce the adverse effects of peristaltic pumps on cells. Additionally, bearingless low-shear centrifugal pumps have been introduced as an alternative to peristaltic pumps to reduce cell lysis and improve product sieving in TFF systems

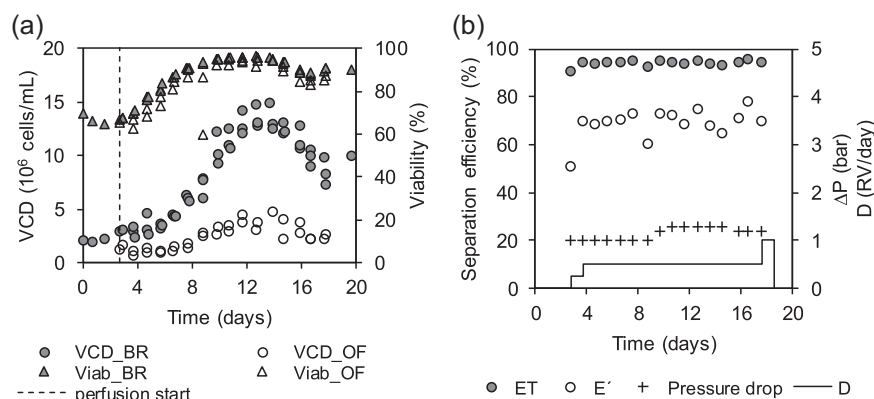


FIGURE 4 Perfusion run #1, started by diluting a late-stage fed-batch CHO culture at 70% viability. A standard 50-L disposable bag with a nonideal 1-m long thick connection to the HC2015 was used. (a) Viable cell density (VCD) and cell viability measured in the bioreactor (BR) and harvest (OF). (b) Perfusion rates (D) and separation efficiencies determined throughout the run (pressure drops between 1.0 and 1.3 bar). CHO, Chinese hamster ovary

(Karst et al., 2016; Wang et al., 2017), and their use could eventually be extended to HC application.

Perfusion run #3 was then designed to gather the best conditions for assembling and operating the HC. The underflow was connected to the bioreactor bag to allow an umbrella-shaped discharge, and a larger ID tubing (3/4") was used to connect the bioreactor to the HC feed inlets. Assuming this operational condition, and due to the unavailability of a single-use flowmeter, Q_f values (shown in Figure 6e) were estimated with an empirical model (see Figure S1). Using a fortified ActiPro composition as feed medium, a maximum cell density of 54×10^6 cells/ml was attained on Day 16 in the bioreactor, with very high viability (Figure 6a), whereas a cell density of 15×10^6 cells/ml was determined in the overflow (Figure 6a). The operational ΔP was maintained at 1.0–1.5 bar until Day 8, and then increased to 2.0–2.5 bar (Figure 6c). LDH levels (Figure 6d) increased with time as did VCD, and were not boosted by pressure drops higher than 2 bar in contrast to perfusion #2, which was carried out with flow limitations due to a smaller diameter of the bag port leading to the HC feed.

Total separation efficiency in run #3 remained at approximately 95% throughout the run. Despite the pressure increase after Day 8, centrifugal efficiencies remained at 75% (Figure 6c), and this could be related to the more concentrated feed suspension in this run. The negative impact of increased cell concentration in the feed suspension on separation efficiency has already been reported for non-membrane based devices (Kwon et al., 2017; Ryll et al., 2000). For example, a decrease in retention efficiency from 99% to 82% was observed in a microfluidic device applied in a 350-ml perfusion system as the cell concentration increased up to 20×10^6 cells/ml (Kwon et al., 2017).

Diluted cell suspensions with lower viability and smaller cell diameter than inside the bioreactor were obtained in the overflow (Figures 5b and 6b). This agrees with Pinto et al. (2008), who observed preferential retention of viable cells with partial elimination of smaller and lighter dead cells through the overflow. Cell retention devices that enable separation of particles with different sizes and densities under the gravitational or centrifugal field, such as the HC,

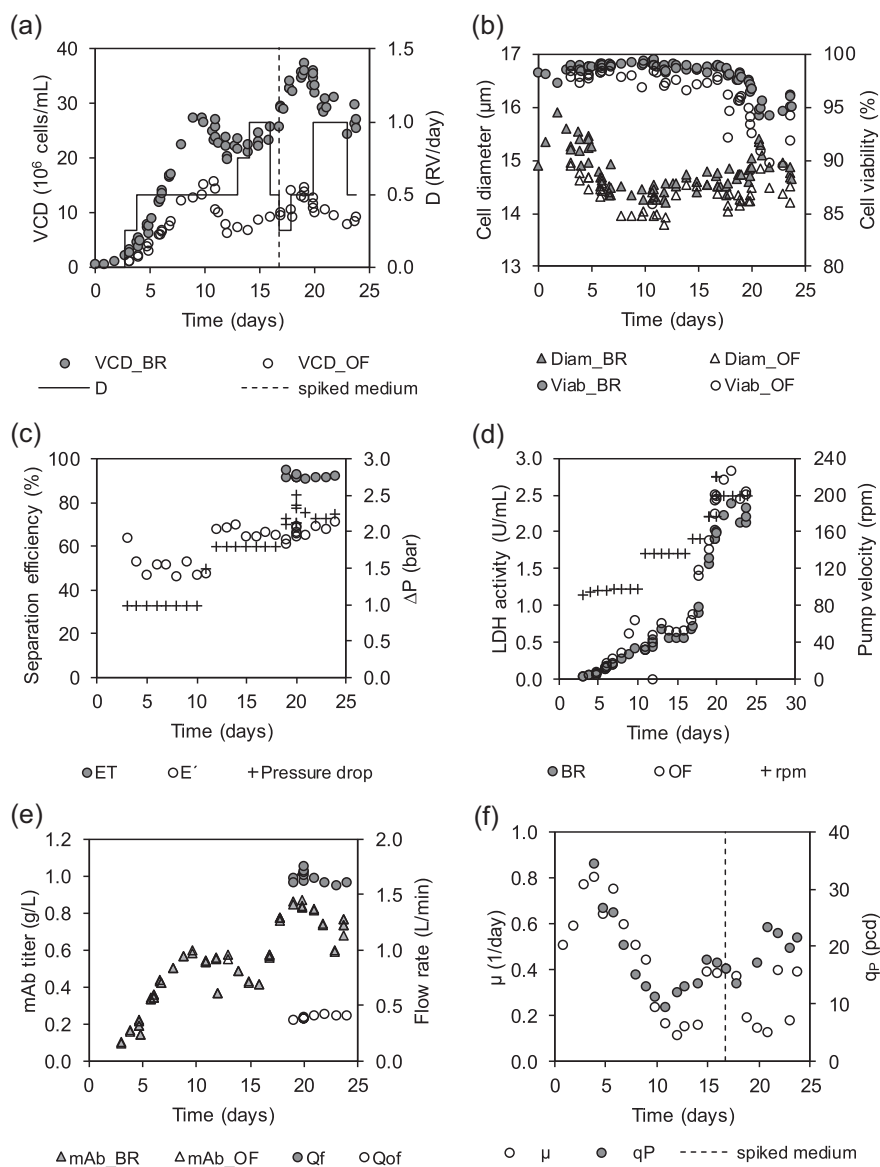


FIGURE 5 Perfusion run #2 carried out in a nonoptimal, customized 1.5" ReadyMate PRO bag with a 3.2-mm ID tubing for the HC feeding. (a) Cell density in the BR and harvest (OF). On Day 18, perfusion rate was reduced to 0.25 RV/day to avoid possible hyperosmotic shock by a spiked feed medium. (b) Cell diameter and viability measured in the bioreactor and harvest. (c) E' and E_T (monitored from Day 19 onwards with a disposable flowmeter). (d) LDH levels in the bioreactor and harvest. (e) IgG concentration measured in bioreactor and harvest, and experimental values of Q_f and Q_{of} (monitored from Day 19 onwards with a disposable flowmeter). Product titers measured in the bioreactor and harvest are overlapped. (f) Specific growth rate (μ) and cell-specific productivity (q_p). BR, bioreactor; HC, hydrocyclone; ID, internal diameter; IgG, immunoglobulin G; LDH, lactate dehydrogenase; Q_f , feed flow rate

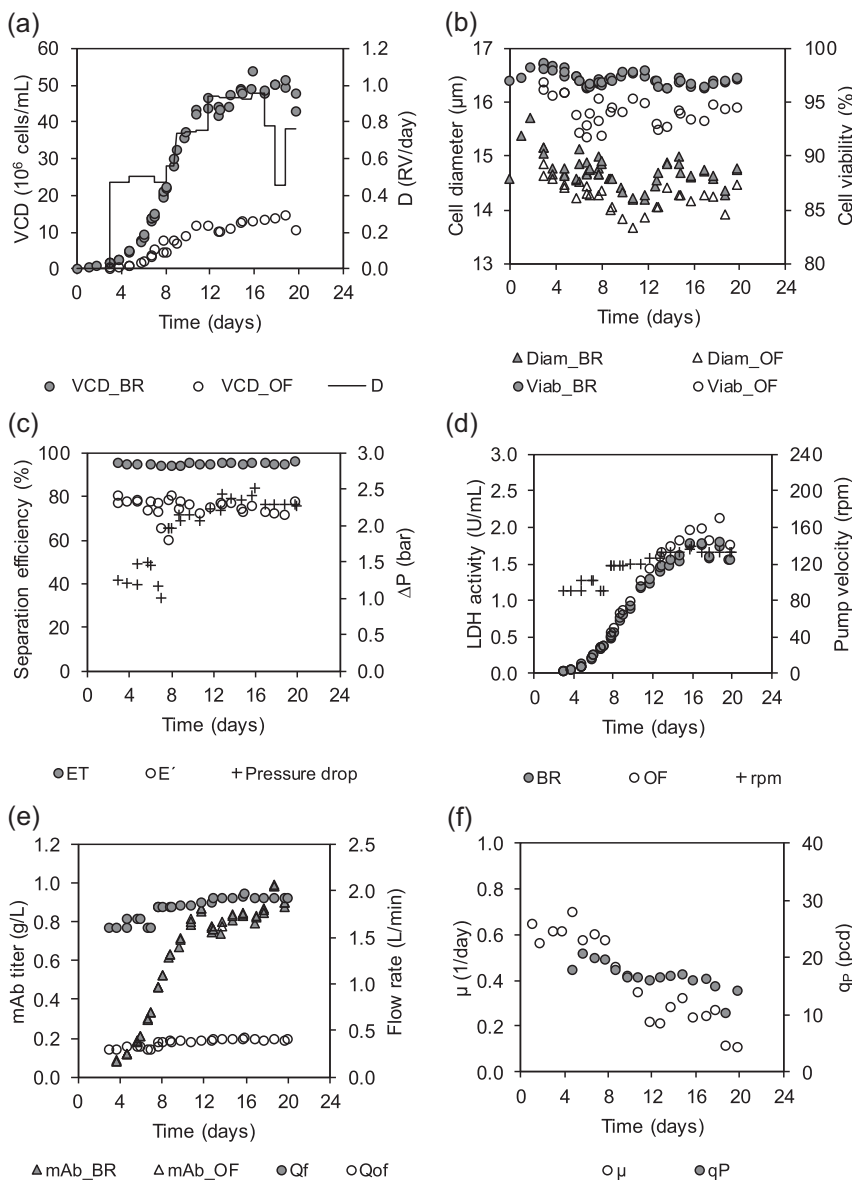


FIGURE 6 Perfusion run #3 carried out in an optimal, customized DEV bag with a 1.5" ReadyMate port and a larger $\frac{3}{4}$ " ID tubing for the HC feed line. (a) Cell density in the BR and harvest (OF). (b) Cell diameter and viability measured in the bioreactor and harvest. (c) E' and E_T . (d) LDH levels in the bioreactor and harvest. (e) IgG concentration measured in bioreactor and harvest (overlapped sequences) and feed flow rates (Q_f and Q_{of}). (f) Specific growth rate (μ) and cell-specific productivity (q_p). BR, bioreactor; HC, hydrocyclone; ID, internal diameter; IgG, immunoglobulin G; LDH, lactate dehydrogenase; Q_f , feed flow rate

are prone to preferentially retain viable cells that present higher density (Batt, Davis, & Kompala, 1990; Patil & Walther, 2017; Takagi, Ilias, & Yoshida, 2000). A natural cell bleeding occurred through the overflow, enabling a healthier culture environment, as had been already observed in run #1, which started from a low-viability late-stage fed-batch culture. On the other hand, having a natural bleed implies that a minimum specific cell growth rate is required to allow high cell concentrations to be achieved and kept through the run, and demands a further clarification step downstream of the HC, as usually required by non-membrane-based cell retention devices. Since HC performance in terms of flow ratio (which determines Q_{of}) and separation efficiency is highly influenced by Q_f (which determines pressure drop and centrifugal field intensity inside the HC), careful optimization of the process needs to take the natural bleed in consideration.

Antibody titers determined in runs #2 and #3, which were carried out with the same CHO cell line, presented a similar

profile to the respective viable cell density profiles, suggesting growth-associated mAb production. In run #2, the cell-specific productivity (q_p) increased after Day 18, that is, after the basal medium was replaced by a fortified formulation that boosted production rates by 70% (Figure 5f), which is in agreement with Reinhart et al. (2018) for the same medium composition. IgG titer was monitored both in the bioreactor and in the harvest, and no product retention was observed, as expected for a HC (Figure 5e,f). This is clearly an advantage over filtration-based cell retention devices.

Due to their high capacity, lack of product retention issues, low cost also when used as disposable devices, and no need for maintenance during operation, HCs are especially suited for pilot- and industrial-scale applications. To enable small-scale process development, different approaches could be adopted. One approach is based on intermittent medium renewal, as adopted in the present work, but adjusting the on-off cycles to match the smaller scale. We have

successfully operated lab-scale perfusion bioreactors with 3–4 L working volume operating the HC intermittently (unpublished results) using a simple socket timer available in the consumer market to automatically switch on feed pump for very short operation times (e.g., 1 min on). Scale-up of processes developed in this way are expected to be simple, since overflow throughput can be increased by shortening the time intervals between HC operation cycles and/or by extending the time that the HC feed pump remains on. In more sophisticated setups, intermittent HC feed pump operation could be driven by a control loop based on the signal of a viable cell density sensor (e.g., capacitance sensor), enabling a steady-state viable cell density to be maintained in the bioreactor.

On the other hand, if the preferred approach is to develop truly continuous perfusion processes, then HCs smaller than the one used in the present work would be needed, since the intensity of the centrifugal field at lower flow rates will just be enough to separate mammalian cells if devices with smaller inner diameter are used. Mini- and micro-HCs described elsewhere (Bhardwaj, Bagdi, & Sen, 2011; Syed et al., 2017; Yu & Fu, 2019), possibly produced by 3D-printing, may enable true continuous operation in scale-down perfusion systems. Scale-up, in this case, would be modular and straightforward, by adding more identical HCs to operate in parallel, possibly using systems composed of batteries of identical mini-HCs to provide the high flow rates needed for large scales, in the same way as well established in other industries, such as the oil industry (Golovanchikov, Novikov, Lamskova, & Filimonov, 2018; Mognon, Bicalho, & Ataíde, 2016).

3.4.1 | Product quality attributes

In spite of the different media used in runs #2 and #3, a rather consistent glycoform distribution was obtained in both runs (Table 2). A prevalence (>70%) of fucosylated forms was confirmed, which is in

agreement with other reports in the literature (Karst et al., 2016; Sanchez-De Melo et al., 2015; Xu & Chen, 2016). In the first days of perfusion operation (Day 7–8 of both runs), the dominant glycoform was non-galactosylated (G0F), whereas in later stages (Day 16–19 of cultivation) the mono-galactosylated G1F glycoform became the most prevalent. Similarly, along both runs, a decreasing trend was observed for the non-galactosylated G0 species and an increasing trend became apparent for the G2F galactosylated glycoform. The redistribution of glycoform abundances over the course of perfusion was also observed in perfusion run using the same cell line and media, despite the different type of bioreactor/cell retention device (Mayrhofer, Castan, & Kunert, 2019). The low specific growth rate observed in later stages of runs #2 and #3 (Figures 5f and 6f), which is a common feature of perfusion processes (Clincke et al., 2013; Gagnon et al., 2018; Wang et al., 2017), might be related to the increase in more fully glycosylated species over time. Ammonia, which has been shown to affect the galactosylation pathway (Hossler, Khattak, & Li, 2009), remained at low levels (<2 mM) throughout both runs (data not shown).

In terms of product aggregation, a very low percentage (0.4–0.8%) of high molecular mass species was observed throughout the cultivations, in contrast to a standard fed-batch process (Table 2). Fragments were also found in low amount (1.7–2.5%), whereas monomers were the prevalent species (96.6–97.8%) throughout both runs (Table 2), which is also in agreement with the study using the same cell line and media, but a different type of bioreactor/cell retention device (Mayrhofer et al., 2019).

As shown in Table 2, when compared with a fed-batch run with the same cell line (Castan et al., unpublished results), the level of monomers is very similar, while aggregation is more enhanced in fed-batch and fragmentation more pronounced in perfusion. Regarding the distribution of N-glycoforms, fed-batch profile is similar to the early stage of perfusion, before the decrease of the specific growth rate and redistribution of glycoform abundances occurred, as discussed above.

TABLE 2 Quality attributes for N-glycosylation and aggregation/fragmentation, measured for samples withdrawn in perfusion runs #2 and #3, compared with a standard fed-batch with the same cell line

		Run #2			Run #3	
		Day 8	Day 16	Day 19	Day 7	Day 17
Fed-batch ^a						
Feed medium		ActiPro		ActiPro+CB7a+CB7b	ActiPro+CB1+CB3	
N-glycoforms (%)						
G0	2.6	3.7	2.6	2.4	3.7	2.8
G0F	52.0	45.6	32.4	31.5	51.5	30.8
G1F	37.6	40.3	49.2	48.5	37.4	48.5
G2F	4.8	6.3	11.0	11.5	5.3	11.7
High mannose (M5/M6)	ND	0.3	0.4	0.8	0.4	0.7
Others	3.0	4.0	4.4	5.4	1.8	5.5
Aggregation (%)						
Monomer	97.4	97.8	97.7	97.4	97.3	96.6
Fragment	ND	1.7	1.8	2.2	2.2	2.5
High molecular mass	2.6	0.5	0.5	0.4	0.5	0.8

Abbreviations: ND, not determined.

^aCastan et al., unpublished results with the same cell line.

The data on critical quality attributes (CQAs), such as glycosylation and presence of aggregates, corroborate the fact that literature does not show an effect on CQAs due solely to the cell retention device employed (Karst et al., 2016; Ryll et al., 2000; Zhang, Stobbe, Orrego, & Chotteau, 2015). This observation is particularly important in the present case, since the HC was intermittently operated with medium exchange performed at several cycles per day, thus representing a pseudoperfusion process.

4 | CONCLUSIONS

In this work, the potential of using HCs as cell retention devices in perfusion cultures carried out in single-use bioreactors was investigated. Initial batch tests allowed determining important operational conditions that were later used in perfusion runs. It was observed that conventional peristaltic pumping, which yields pulsating flow, can be used to feed HCs. Placing a peristaltic pump in the overflow line to control the perfusate flow rate affected the flow inside the HC and decreased centrifugal separation efficiency. Regarding the connection of the HC to a single-use bioreactor bag, the importance of having a large-diameter recirculation loop to connect the bag to the HC feed orifices, and the HC underflow back to the bag, was demonstrated by testing tubings and aseptic connectors of different diameters.

Perfusion runs were then carried out to evaluate the feasibility of HC intermittent operation connected to a 50-L single-use bioreactor. The first perfusion run was a pilot experiment that was started by diluting a low-viability late-stage fed-batch culture and allowed recovering cell viability from about 70% to more than 95%. Perfusion run #2 was 24 days long and reached approximately 30×10^6 cells/ml, but used a regular bioreactor bag with conventional ports that prevented HC optimal operation due to the small diameter of the lines leading to the HC double-feed entrance. In perfusion run #3 a customized bioreactor bag with large-diameter ports allowed having an adequate diameter in the recirculation loop, that is, in the bioreactor bag connection to the HC feed and returning from the HC underflow orifice. This run was carried out for 20 days and reached a steady-state at approximately 50×10^6 cells/ml, which is five-fold higher than the maximum viable cell density that had been published so far for a perfusion process based on a HC as cell retention device (Elsayed, 2005). Total separation efficiencies of 95% and centrifugal separation efficiencies of 70–80% were obtained throughout the run. Preferential removal of dead cells was confirmed in the overflow (harvest) line.

HCs, such as settlers, centrifuges, and acoustic separators, are cell retention devices that do not use a barrier to promote separation. As such, the progressive clogging inherent to barrier-based devices, such as ATF and TFF, is not an issue, and this was confirmed in the present work since mAb concentrations in the harvest and inside the bioreactor were equal, that is, no issues of product retention were experienced.

Similarly to other devices with no barrier to promote separation, as shown in the present work, overall cell separation efficiencies in the HC can be very high but do provide a natural bleed. This couples

dilution rate and separation efficiency, but on the other hand contributes to maintaining a healthier culture in the bioreactor.

The lack of moving parts makes the HC a simple and reliable device, decreasing CAPEX and OPEX costs. Although HCs had been already tested for perfusion cultures in the past, previous reports were mainly restricted to rather low cell densities determined by the limitations of the protein-free media used by then. This is thus the first report to show that HCs can stably sustain steady-state perfusion in the range of 50×10^6 cells/ml. Further studies with this HC are ongoing, including investigation of intermittent perfusion at smaller laboratory scale and at higher cell densities.

HCs are very sensitive to the inner geometrical proportions, so a careful HC manufacturing process is critical. 3-D printing can streamline the manufacturing of HCs for single-use applications, as well as enable manufacturing of HCs even smaller than the one used in the present work, which is of interest because the smaller the HC is (as given by the internal diameter of its cylindrical body), the higher the centrifugal separation efficiency is.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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